WEATHER DATA ANALYSIS AND PREDICTION PROJECT

Weather forecasting requires processing huge volumes of meteorological data collected from satellites, weather stations, radar systems, and sensors. Traditional systems cannot efficiently process such massive datasets.The Weather Data Analysis and Prediction Project uses Big Data technologies, especially the Apache Hadoop ecosystem, to store and analyze large-scale weather data. By using distributed computing and parallel processing, the system can quickly analyze weather patterns and generate predictions.This project involves analyzing historical weather data (temperature, humidity, pressure, precipitation) using Python (Pandas, Matplotlib) and machine learning algorithms (Linear Regression, Naive Bayes) to identify trends and predict future conditions. Key stages include data cleaning, exploratory data analysis (EDA), and visualization to provide actionable insights into climatic patterns

Key Project Components & Technologies

- **Data Analysis & Visualization:** Use Pandas for data manipulation, and Matplotlib or Seaborn to visualize temperature trends, humidity, and correlation heatmaps.
- **Machine Learning Prediction:** Implementation of regression models (e.g., Linear Regression) to predict future temperature or classification models to forecast weather conditions.
- **Tools Used:** Python, Jupyter Notebook, SQL (MySQL) for data management, and libraries like Scikit-learn for modeling.
- **Key Metrics Analyzed:** Daily maximum/minimum temperature (TMAX, TMIN), precipitation (PRCP), dew point, and humidity.
- **Project Output:** An interactive dashboard (Power BI/Tableau) or a Python-based model that projects future 30-day weather trends

Steps in the Project

- **Data Collection:** Gathering historical datasets (e.g., CSV files, weather APIs).
- **Data Preprocessing:** Handling missing values, converting date formats, and cleaning data.
- **Exploratory Data Analysis (EDA):** Identifying relationships between variables (e.g., Temperature vs. Humidity).
- **Model Building:** Training algorithms like Linear Regression from Sathyabama or SVM on historical data.
- **Prediction and Visualization:** Forecasting future trends and plotting results.